INCH-POUND

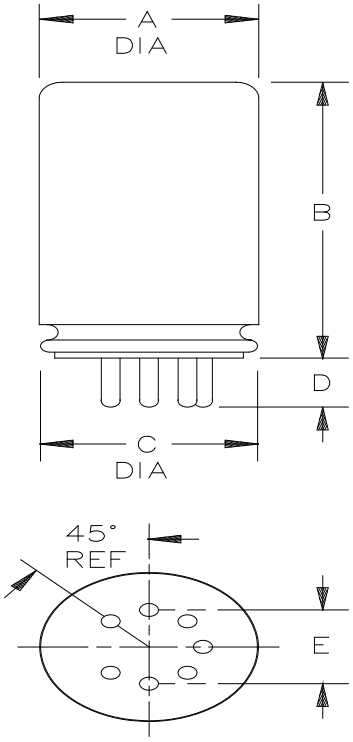
MIL-DTL-83729/6C
13 March 2013
SUPERSEDING
MIL-DTL-83729/6B
14 June 2002

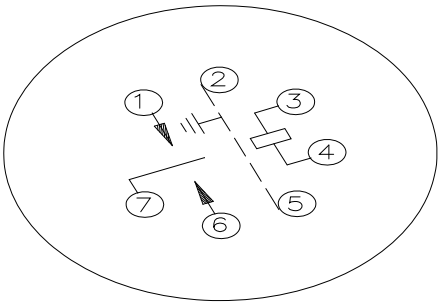
DETAIL SPECIFICATION SHEET

CHOPPERS, ELECTROMECHANICAL, HERMETICALLY SEALED, SINGLE POLE, DOUBLE THROW, BREAK BEFORE MAKE

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83729.





SCHEMATIC DIAGRAM

Letter	Inches	mm
A	.763 MAX	19.38 MAX
B	1.438	36.53
B	1.391	35.33
C	.763	19.38
C	.745	18.92
D	.265	6.73
D	.235	5.97
E	.375	9.53

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerances is ± .005 (.13 mm).
4. Manner of presenting the diagram is optional. Numbers are not required if shown with respect to terminal arrangement.
5. When the coil is de-energized the center reed may be in any of 3 positions.

FIGURE 1. Outline drawing, terminal arrangement and schematic diagram.

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REQUIREMENTS:

Contact data:

Single pole, double throw, break before make.
Polarity: With positive dc voltage on pin 3 and negative voltage on pin 4, pins 6 and 7 are closed.
Voltage and current: Dry circuit to 100 volts, 2 milliamperes dc resistive (maximum intermittent rating) 10 volts, 1 milliampere continuous dc resistive rating.

Coil data:

Voltage rated: 6.3 volts ± 0.6 volts rms.
Resistance, dc: 155 ohms ± 10 percent at 25°C.
Impedance: See table I.
Frequency range: See table I.

Waveform characteristics (6.3 volts at 25°C ± 5°C):

Dwell time: See table I.
Dissymmetry: 15° maximum.
Transfer time: See table I.
Phase angle: See table I.

Noise level: See table I.

Environmental conditions:

Dielectric withstanding voltage: See table I.
Insulation resistance: Between terminals and case 1,000 megohms at 100 volts dc.
Vibration III: Nonoperating, MIL-STD-202, method 204, condition A, except 5 g's maximum.
Shock: Nonoperating, 100 g's as specified in MIL-STD-202, method 213, test condition I.
Temperature range: -65°C to +125°C.

Life: 5,000 hours minimum at dry circuit, 25°C ± 5°C.

Mechanical:

Weight: .09 pound, maximum.
Dimensions: See figure 1.
Enclosure: Hermetically sealed, chopper mates with JAN TS102P01 socket with a TS102U02 tube shield to secure in the socket.
Terminal strength: 2.5 pounds.
Mounting position: Any.

Part or Identifying Number (PIN): M83729/6- (dash number from table I).

TABLE I. Dash number and characteristics.

Dash number	Imped-ance	Frequency range	Dwell time	Transfer time	Phase angle	Noise level	Dielectric Withstanding voltage
001	250 ± 20% at 25°C	400 ± 20 Hz	130° to 165°	15° min	65° ± 15°	200 microvolts rms above system noise level with 1.0 meg-ohm load resistor.	200 V rms at 60 Hz or 280 V dc
002	175 ± 20% at 25°C	60 ± 5%	150° min	3° min	20° ± 5°	50 microvolts rms above system noise level with 1.0 meg-ohm load resistor.	200 V rms at 60 Hz or 280 V dc

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Referenced documents. In addition to MIL-DTL-83729, this document references the following:
MIL-STD-202

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Preparing activity:
DLA – CC

(Project 5945-2013-015)

Review activities:
Army – AT, AV, MI
Navy – AS, MC, OS
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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